

Which Block-Rotation Puzzles Generate the Symmetric Group?

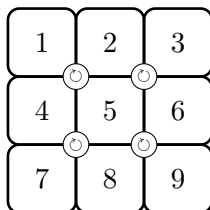
Michael Albert
University of Otago

Abstract

For an $m \times n$ grid of cells, let $G_{m,n}$ be the group generated by clockwise 90° rotations of all 2×2 blocks. We give an elementary, self-contained proof that, for $2 \leq m \leq n$, $G_{m,n}$ is the full symmetric group on the mn cells if and only if $m \geq 3$ or $n \geq 4$ — avoiding classical primitivity arguments such as Jordan’s theorem in favour of direct computations, the grid’s own geometric symmetries, and a short exhaustive search. As a by-product, we show that the one interesting exceptional case, $G_{2,3}$, is isomorphic to S_5 , realizing on this puzzle-derived set of 6 points the classical exotic degree-6 representation of S_5 .

1 Introduction

Consider this grid



and the question “Which permutations of its cells can be achieved by rotating the four 2×2 corner blocks repeatedly?”

More formally, let $G \leq S_9$ be the group generated by the four elements:

$$a = (1\,2\,5\,4), \quad b = (2\,3\,6\,5), \quad c = (4\,5\,8\,7), \quad d = (5\,6\,9\,8),$$

with the goal being to understand the structure of G .¹

The corresponding group arose in a puzzle that I was working on in the 4×4 case, which led me to this exploration. The result is not new — in fact, a more complete treatment can already be found in a paper by Lam [6] — but the argument here is elementary, different, and self-contained.

First we’ll show that G contains a 3-cycle. That’s a standard step in “block moving puzzles”. From that point, I could blind you with science [2]: *it is easy to see that G is 2-transitive, hence primitive, so that Jordan’s theorem [1, 5] implies that G is either A_9 or S_9 ; but the generators are odd, so only the latter is possible.*

However, in the spirit of the puzzle that inspired this note in the first place, we’ll give a more elementary argument that proves that *all* 3-cycles belong to G .

¹Spoilers: G is the full symmetric group S_9 , and the same holds for $m \times n$ grids for all $m, n \geq 3$ and also for $2 \times n$ strips for $n \geq 4$.

2 A symmetry observation

The 3×3 grid carries an obvious geometric symmetry: the dihedral group D_4 of order 8 (the four rotations and four reflections of the square) acts on the 9 cells, and this action permutes the four 2×2 corner blocks among themselves. It therefore also acts, by conjugation, on the generating set $\{a, b, c, d\}$ — and a direct check shows this action follows the geometry exactly:

- the rotations of the grid permute a, b, c, d amongst themselves, following the blocks around the centre: a 90° clockwise rotation of the whole grid sends $a \mapsto b \mapsto d \mapsto c \mapsto a$, and a 180° rotation swaps $a \leftrightarrow d$ and $b \leftrightarrow c$;
- the four reflections send each generator to the *inverse* of a generator, e.g. the reflection across the main diagonal fixes the blocks of a and d setwise but sends $a \mapsto a^{-1}$, $d \mapsto d^{-1}$, and swaps $b \mapsto c^{-1}$, $c \mapsto b^{-1}$.

The pattern is exactly orientation: an orientation-preserving symmetry of the plane turns a clockwise rotation of a block into a clockwise rotation of the image block, i.e. another generator, while an orientation-reversing symmetry turns a clockwise rotation into a counter-clockwise one, i.e. the inverse of a generator. In particular every element of D_4 conjugates $\{a, b, c, d\}$ back into $\{a^{\pm 1}, b^{\pm 1}, c^{\pm 1}, d^{\pm 1}\} \subset G$, so D_4 normalizes G .

This lets us transfer any fact proved about the action on one collection of cells to its whole D_4 -orbit for free, without repeating the underlying computation.

3 Creating all 3-cycles

In generating all 3-cycles it suffices to generate a 3-cycle of each possible support (the set of three points that it moves) since, if two 3-cycles have the same support, each is the square of the other. So the plan is to transport supports of 3-cycles around the grid using both conjugation by elements of the group itself, and the symmetry of the grid. We begin with a direct calculation that produces a 3-cycle from the generators.

$$a^{-1}d^{-1}ad = (4\ 5\ 6).$$

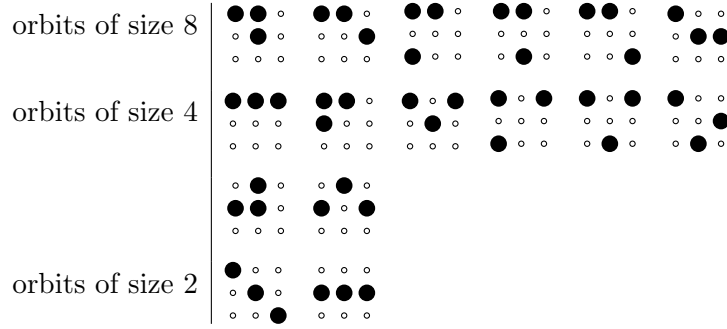
See Appendix A for a lemma that provides hints about where to hunt for initial 3-cycles in puzzles of this type.

Beginning from our single 3-cycle $(4\ 5\ 6)$, we will successively construct more 3-cycles with different supports until we have one for every symmetry type of a triple of distinct cells.

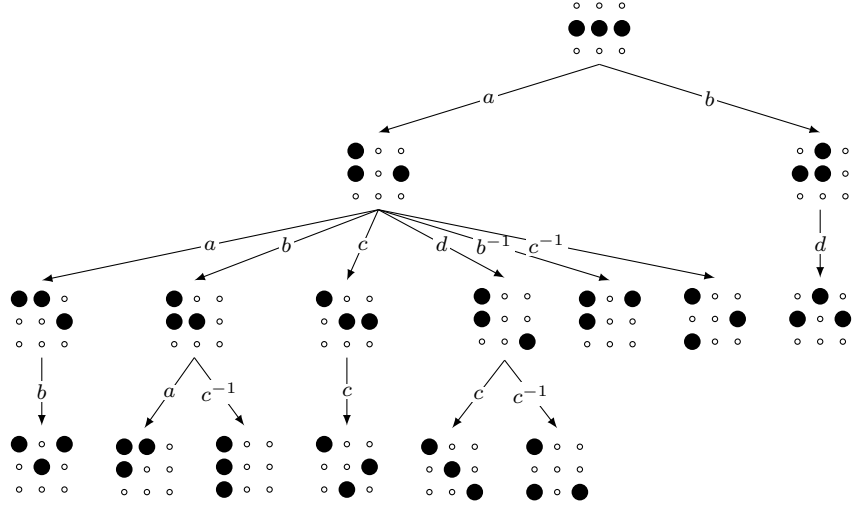
For this purpose we use a simple graphical representation of 3-element subsets of the 3×3 grid — for instance, the support of the 3-cycle $(4\ 5\ 6)$ is represented as:



There are $\binom{9}{3} = 84$ 3-element subsets of the grid. A direct check shows that up to the symmetry of D_4 , there are exactly 16 orbits: six of size 8, eight of size 4, and two of size 2. One representative of each orbit is shown in the following table:



The last of these, $\{4, 5, 6\}$, is the support we already have. The task is to find, for each of the remaining 15 shapes a way to reach it from $\{4, 5, 6\}$ by conjugation with elements of G . The tree below shows the results of such a search.



A representative of every one of the 16 shapes from the catalog appears exactly once in this tree, so G contains a 3-cycle of every possible support and, as noted above, this is enough to conclude that G contains every 3-cycle in S_9 .

The 3-cycles generate A_9 , so we now know that $A_9 \leq G$. The generator $a = (1\ 2\ 5\ 4)$ is a 4-cycle, hence an odd permutation, so $a \notin A_9$. Since A_9 is a maximal subgroup of S_9 ,

$$G = S_9. \quad \blacksquare$$

4 Generalization to $m \times n$ grids

The argument extends to non-square grids, as long as both dimensions are at least 3.

Theorem 1. *Let $m, n \geq 3$, and let $G_{m,n} \leq S_{mn}$ be generated by the $(m-1)(n-1)$ rotations of the 2×2 blocks of an $m \times n$ grid. Then $G_{m,n} = S_{mn}$.*

Proof. Any 3×3 block of consecutive rows and columns is itself a 3×3 grid whose four corner-rotation generators are among the generators of $G_{m,n}$; by the previous results, that block's local

group is the full symmetric group on its 9 cells, fixing everything outside it. Every grid-adjacent pair of cells lies inside some such 3×3 block so $G_{m,n}$ contains every adjacent transposition. The grid's adjacency graph is connected, and transpositions indexed by the edges of such a graph generate the full symmetric group on its vertices, so $G_{m,n} = S_{mn}$. $\square$

5 The $2 \times n$ strip

For completeness, consider the case $m = 2$, a grid with two rows. Label the $2 \times n$ grid with cells $1, \dots, n$ in the top row, and $n+1, \dots, 2n$ in the bottom row, and let $G_n \leq S_{2n}$ be generated by the $n - 1$ block rotations

$$g_i = (i \ i+1 \ i+n+1 \ i+n), \quad i = 1, \dots, n - 1.$$

$n = 2$: Too small to matter

$G_2 = \langle g_1 \rangle$ is cyclic of order 4, far short of $|S_4| = 24$.

$n = 3$: Proper, by direct computation

Direct computation gives $|G_3| = 120 < 720 = |S_6|$, so $G_3 \neq S_6$. In fact G_3 is isomorphic to S_5 , as is proved in Appendix B.

$n = 4$: Full, via symmetry and a short search

Adjacent generators of the strip overlap in *two* points, not one, so Lemma 2 does not apply to g_1, g_2 to give a 3-cycle directly. However,

Lemma 1. G_4 contains the 3-cycle $(2 \ 3 \ 4)$.

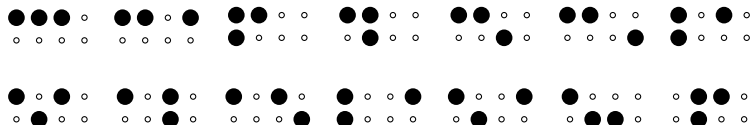
Proof. A direct check of the product gives

$$[g_1, g_2] = (2 \ 3)(5 \ 6),$$

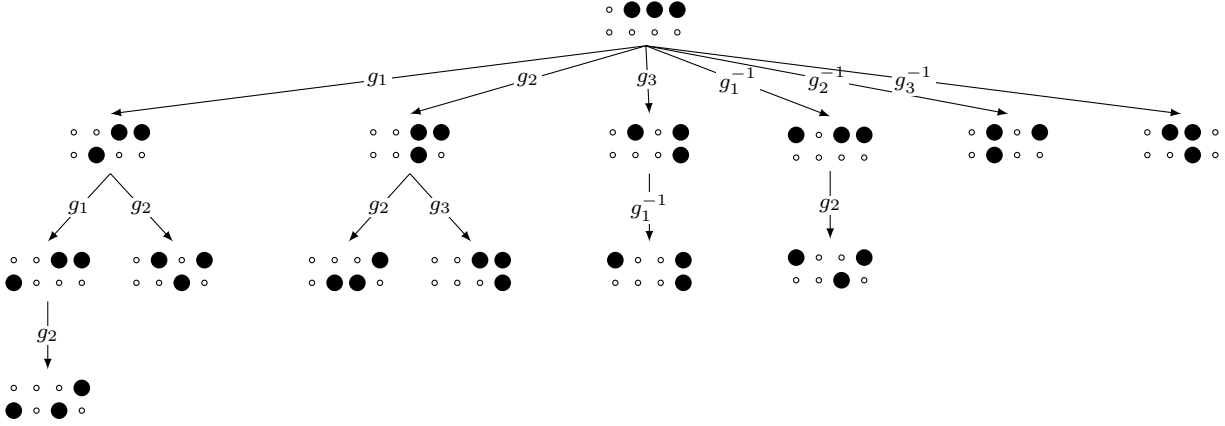
a double transposition, and this *does* share exactly one moved point (3) with g_3 . Lemma 2 applies to $\sigma = [g_1, g_2]$, $\tau = g_3$ giving $[\sigma, \tau] = (2 \ 3 \ 4)$. $\square$

The 2×4 strip is not square, so it has no 90° rotational symmetry — only the Klein four-group $V_4 = \{e, h, v, hv\}$ of order 4, where h is the left-right mirror and v is the top-bottom mirror of the strip. But we are free to use this more limited group of symmetries to sweep the 3-cycle $(2 \ 3 \ 4)$ around the strip.

So first we need to find representatives of all the possible triples. Here, each orbit has size four since none of the non-identity elements of V_4 can fix a 3-element subset. Since $\binom{8}{3} = 56 = 14 \times 4$, there are exactly 14 orbits. One representative of each is shown below:



As in the 3×3 case, we now do a breadth-first search on V_4 -symmetry-classes of triples, starting from $\{2, 3, 4\}$. A direct search reaches all 14 shapes from the catalog above by depth 3:



Every one of the 14 shapes appears exactly once in this tree, so G_4 contains a 3-cycle of every possible support, hence every 3-cycle in S_8 . As before this gives $G_4 \geq A_8$; the generator g_1 is a 4-cycle, hence odd, so $G_4 \not\leq A_8$, and since A_8 has index 2 in S_8 ,

$$G_4 = S_8.$$

$n \geq 5$: The same window argument as before

Theorem 2. $G_n = S_{2n}$ for every $n \geq 4$.

Proof. As in Theorem 1: any four consecutive columns form a 2×4 strip whose local generators (three of G_n 's generators) generate the full symmetric group on those 8 cells, by the $n = 4$ case just proved. Every grid-adjacent pair of cells lies inside some such window once $n \geq 4$, so G_n contains every adjacent transposition; the grid graph is connected, so $G_n = S_{2n}$. $\square$

6 The complete answer

Theorem 3. Let $2 \leq m \leq n$, and let $G_{m,n}$ be the group generated by all 2×2 cell rotations of an $m \times n$ grid. Then

$$G_{m,n} = S_{mn} \iff m \geq 3 \text{ or } n \geq 4.$$

Proof. If $m \geq 3$ (so also $n \geq 3$), this is Theorem 1. If $m = 2$, this is Theorem 2 together with the cases $n = 2, 3$ considered above. $\square$

Acknowledgements

I thank Claude (Anthropic), who provided extensive assistance throughout this project: verifying every computational claim, helping develop the symmetry-and-search arguments that replaced an

earlier, less illuminating approach, tracking down the literature cited here, and copy-editing the manuscript.

The puzzles Tintangle and Wordtangle that inspired this investigation can be accessed at:

<https://michael-albert-dun.github.io/>

Appendix A A lemma on single-point overlaps

The following lemma (or minor variations of it) is a standard fact about permutations. It plays exactly this role in the usual proof that a primitive group containing a 3-cycle contains the alternating group [1]. It also appears under the same guise (“the commutator of two moves is a 3-cycle”) in the theory of puzzle groups on graphs [7], but we include a proof for completeness.

We use exponential notation for the action of permutations: for $x \in \Omega$ and $\sigma \in S_\Omega$, write x^σ (rather than the more standard $\sigma(x)$) for the image of x under σ . Recall that the support of a permutation is the set of points it moves.

Lemma 2. *Let $\sigma, \tau \in S_\Omega$ be permutations whose supports intersect in exactly one point x . Set $y = x^\sigma$ and $z = x^\tau$, then*

$$[\sigma, \tau] = \sigma^{-1}\tau^{-1}\sigma\tau = (y \ x \ z).$$

Proof. Let S be the support of σ and T be the support of τ . Then $S \cap T = \{x\}$. Any point outside $S \cup T$ is fixed by all four factors $\sigma^{-1}, \tau^{-1}, \sigma, \tau$, hence fixed by $[\sigma, \tau]$. Note that $y \in S$, $z \in T$ and neither is equal to x , hence $y \neq z$, $y \notin T$ and $z \notin S$. Thus $y^\tau = y$ and $z^\sigma = z$.

We claim that any element not in $\{x, y, z\}$ is fixed by $[\sigma, \tau]$. This is already established for elements outside $S \cup T$. Suppose $u \in S \setminus T$ with $u \neq y$. Then $u^{\sigma^{-1}} \neq x$, for otherwise $u = x^\sigma = y$; since σ^{-1} preserves S , this places $u^{\sigma^{-1}}$ in $S \setminus \{x\} \subseteq S \setminus T$, so it is fixed by τ^{-1} , and applying σ returns u ; finally τ fixes u since $u \notin T$. Hence $u^{[\sigma, \tau]} = u$. Similarly, any $u \in T \setminus S$ with $u \neq z$ is fixed by $[\sigma, \tau]$. As $S \cap T = \{x\}$, this accounts for every point outside $\{x, y, z\}$.

It remains to track x, y, z . Since $x^{\sigma^{-1}} \in S \setminus \{x\} \subseteq S \setminus T$ (as above), it is fixed by τ^{-1} , so

$$x^{[\sigma, \tau]} = x^{\sigma^{-1}\tau^{-1}\sigma\tau} = x^{\sigma^{-1}\sigma\tau} = x^\tau = z.$$

Likewise $y^{\sigma^{-1}} = x$, and $x^{\tau^{-1}} \in T \setminus \{x\} \subseteq T \setminus S$ is fixed by σ , so

$$y^{[\sigma, \tau]} = x^{\tau^{-1}\sigma\tau} = x^{\tau^{-1}\tau} = x.$$

Since $[\sigma, \tau]$ fixes everything outside $\{x, y, z\}$ and sends $x \mapsto z$, $y \mapsto x$, the remaining value $z^{[\sigma, \tau]}$ is forced to be y , so $[\sigma, \tau]$ is the 3-cycle $(y \ x \ z)$. $\square$

Appendix B $G_3 \cong S_5$

The action of G_3 is *primitive* on its 6 cells — there is no invariant partition to blame. What explains the shortfall $|G_3| = 120 < 720 = |S_6|$ is that G_3 contains no 3-cycle at all (its order-3 elements are all products of two disjoint 3-cycles), and its conjugacy class sizes (1, 10, 15, 20, 30, 24, 20, for orders 1, 2, 2, 3, 4, 5, 6) match those of S_5 exactly. It is a classical fact that S_5 has an “exotic”

transitive representation of degree 6 — via the exceptional isomorphism $S_5 \cong \text{PGL}(2, 5)$ acting on the 6 points of the projective line over $\mathbb{F}_5$ — distinct from its natural degree-5 action [4]. We now confirm that G_3 really is this representation.

Theorem 4. $G_3 \cong S_5$.

Proof. Direct computation gives the following facts about $G_3 = \langle g_1, g_2 \rangle$ (order 120, on 6 cells).

The centre is trivial. No non-identity element commutes with both g_1 and g_2 .

The derived subgroup $D = G'_3$ has order 60, hence index 2 in G_3 , thus is normal.

D is simple. Its conjugacy classes (inside D) have sizes 1, 12, 12, 15, 20, of elements of orders 1, 5, 5, 2, 3 respectively — already the class structure of A_5 (one class of double transpositions, one of 3-cycles, and the classic pair of 5-cycle classes that fuse in S_5 but split in A_5). Any normal subgroup of D is a union of conjugacy classes including the identity's, with order dividing 60; checking all subsets of $\{12, 12, 15, 20\}$, none of the resulting sixteen sums plus 1 is a proper nontrivial divisor of 60. So D has no nontrivial proper normal subgroup.

Since A_5 is, up to isomorphism, the unique simple group of order 60 [3], $D \cong A_5$.

G_3 acts on D by conjugation, since $D \trianglelefteq G_3$; this gives a homomorphism $\varphi: G_3 \rightarrow \text{Aut}(D) \cong \text{Aut}(A_5) \cong S_5$ (the last isomorphism is again classical: $\text{Aut}(A_n) \cong S_n$ for $n \neq 6$). As D is nonabelian simple, $Z(D) = 1$, so $\ker \varphi \cap D = Z(D) = 1$; since D has index 2, this forces $|\ker \varphi| \leq 2$. If $|\ker \varphi| = 2$, its unique nontrivial element would generate a normal subgroup of order 2 in G_3 — but the unique nontrivial element of a normal order-2 subgroup is fixed by every conjugation, i.e. lies in $Z(G_3)$, contradicting $Z(G_3) = 1$. So $\ker \varphi = 1$: φ is injective. Since $|G_3| = 120 = |S_5|$, φ is an isomorphism. $\square$

So G_3 sits inside S_6 , as an index-6 subgroup, exactly the exotic degree-6 representation of S_5 .

References

- [1] J. D. Dixon and B. Mortimer, *Permutation Groups*, Graduate Texts in Mathematics 163, Springer-Verlag, 1996.
- [2] T. Dolby, *She Blinded Me with Science*, Venice in Peril Records / Capitol Records, 1982. <https://www.youtube.com/watch?v=V83JR2IoI8k>.
- [3] D. S. Dummit and R. M. Foote, *Abstract Algebra*, 3rd edition, John Wiley & Sons, 2004.
- [4] G. J. Janusz and J. Rotman, *Outer Automorphisms of S_6* , American Mathematical Monthly **89** (1982), no. 6, 407–410.
- [5] C. Jordan, *Théorèmes sur les groupes primitifs*, Journal de Mathématiques Pures et Appliquées, 2e série, tome 17 (1873), 351–367.
- [6] T. Lam, *Solution to the Number Rotation Puzzle*, arXiv:2211.16787 [math.CO], 2022.
- [7] R. M. Wilson, *Graph puzzles, homotopy, and the alternating group*, Journal of Combinatorial Theory, Series B **16** (1974), no. 1, 86–96.